UbiMaze: A New Genre of Virtual Reality Game Based on Mobile Devices

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Abstract
Most of current virtual/augmented reality games focus on players' immersive experience. The creative process of game design is usually not accessible to the player. In this paper, we present a new game genre that combines player initiated game design with game play. This new genre enables users to design games in a physical space and then play in a rendered virtual space. To this aim we illustrate our conceptual design of a virtual reality game, called UbiMaze, which promotes player participation, and provides a rich, interactive and engaging experience.

Author Keywords
Mobile 3D; Head-mounted Displays; virtual reality; natural interaction

ACM Classification Keywords
H.5.1. [Information interfaces and presentation]: Multimedia Information Systems - Artificial, augmented, and virtual realities.

Background and motivation
Through the enormous success of Oculus Rift in 2012, virtual reality (VR) is attracting the headlines again. This leads to the development of head-mounted displays (HMDs). Many new HMDs are emerging in the market. Most of them, such as Oculus Rift [4] and

Figure 1: Design a maze structure on the floor

Figure 2: The captured pattern

Figure 3: Render the corresponding 3D maze on the mobile phone

HoloLens [2], contain stereoscopic displays and tracking systems, allowing users to see 3D images and navigate through virtual worlds by physically walking, turning one's body, or looking around. In recent years, the smartphone has become popular and ubiquitous. It can also be used as a mini computer integrated with a multi-touch screen. Given that a smartphone includes computing, storage, input and output functions, cardboard 3D glasses with smartphone such as Google Cardboard [1] and Samsung Gear VR [5] have become another popular type of HMDs. Consequently, more and more researchers and engineers have begun to pay close attention to research and development of the application of cardboard 3D glasses [3].

In existing VR games, models of all the buildings and surroundings in the virtual space generally have been designed in advance. Users are only audience but not storytellers in the virtual environment (VE). A VE must be created manually by using professional 3D modeling tools, such as Maya or 3Ds Max. The modeling process is not only time consuming but also couldn't be done by non-professional users. The 3D modeling tools can seldom be used on smartphones. Recently, a few easy modeling systems have started to appear. For instance, Autodesk 123D [6] allows users to create a VE with low learning overhead. Minecraft [7], a game of placing blocks in VE to build arbitrary environments, is popularly used by many American schools to cultivate students' creativity. However the time cost in building 3D scenes using these kinds of systems is still unbearable which dramatically diminishes the playing interest.

The motivation of our current study is to address the problems of how to (1) construct a VR system that can offer general users a fast, flexible and creative platform for designing a VE, and (2) allow users to move through VEs in as similar a manner as possible to walking through the real world using a smartphone. As a first exploration in this direction, we developed UbiMaze, an interactive maze game as our inspiration.

UbiMaze: a new genre for virtual reality games

UbiMaze has two interaction modes: a "Design" mode, in which a maze can be created by non-technologists; and a "Play" mode, in which the virtual maze can be played anywhere by users without any knowledge of tracking.

Design Mode

In the Design mode, users can use low cost marking tools, such as chalks and ribbons, to create different kinds of maze along the floors. Figure 1 is an example of authoring a VR game by physically applying tapes to the floor.

After designing a maze in the physical space, our system provides an interactive interface for generating the corresponding virtual maze. We first capture the pattern of the designed maze using the smartphone camera. Second, we calibrate the captured pattern using transformation (see Figure 2). Lastly, we mark the maze features, namely the positions of walls, via an intuitive touch interface. The features are then stored in a file and rendered in a 3D model of the scene on the mobile phone (see Figure 3).

Play Mode

After the maze is created, users can switch to the Play Mode.

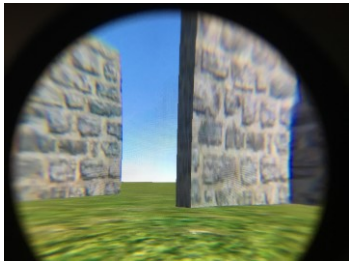

Figure 4: View the virtual scene through cardboard 3D glasses

Figure 5: Play in the maze

The system loads the structure information of the maze and activates the tracking sensor to complete the initialization. The users wear cardboard 3D glasses to view the virtual 3D scene (see Figure 4) and interact with the virtual objects guided by the tracking sensors (see Figure 5).

Conclusion

We presented UbiMaze, a new game genre that fluidly moves users from designing in physical space to playing in virtual ones. UbiMaze enables non-technologists a novel genre in which fast, flexible and creative making of VR games are made possible. UbiMaze allows the easy creation of 2D plan and real-time 3D rendering of VR games. It offers an environment where casual users can easily and quickly immerse themselves in magical worlds.

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